

Georgia Institute of Technology



Thermal Protection Systems and Aerodynamic Heating Analysis

Stagnation Point Heating, Aerodynamic Heating, and TPS Sizing

AE 4080 Computer Project Final Report

Abstract. This project evaluates the thermal environment of a hypersonic projectile by computing stagnation point heating, aerodynamic heating at selected midbody locations, and thermal protection system sizing requirements. Trajectory data were processed using atmospheric calculations, shock models, Fay-Riddell and Sutton-Graves heating correlations, reference temperature methods, and ablative TPS finite difference analysis for Carbon-Carbon and fake PICA material systems.

Author: Brent Perdew

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Contents

Introduction	1
1 Methodology	2
1.1 Stagnation Point Heating	2
1.2 Aerodynamic Heating	3
1.3 TPS Sizing	3
2 Results and Discussion	5
2.1 Stagnation Point Heating	5
2.2 Aerodynamic Heating	7
2.3 TPS Sizing	10
Appendix A: Instructions on Running Code	11

List of Figures

2.1	Stagnation point heat flux as a function of time.	5
2.2	Stagnation point heat load as a function of time.....	6
2.3	Aerodynamic heat flux at reference heating Point 1 as a function of time.....	7
2.4	Aerodynamic heat load at reference heating Point 1 as a function of time.	8
2.5	Aerodynamic heat flux at reference heating Point 2 as a function of time.....	9
2.6	Aerodynamic heat load at reference heating Point 2 as a function of time.	9

List of Tables

2.1	Carbon-Carbon TPS Sizing Results.....	10
2.2	Fake PICA TPS Sizing Results	10

Introduction

This project examines the heating environment experienced by a hypersonic vehicle as it follows a high-energy trajectory under scramjet power before descending into a steep terminal dive. The combined bi-parabolic nose and truncated elliptic cone body create a complex thermal environment in which heat loads vary substantially across the surface as the vehicle travels at significantly high Mach number, through changing density, and while withstanding strong shock formation.

The primary focus of this work is to determine the stagnation point heating throughout the trajectory, evaluate the aerodynamic heating at determined midbody locations along the top and side surfaces, and establish the thermal loads that will later be used for sizing two different thermal protection systems. Accurately predicting these heating levels is essential, as hypersonic vehicles experience severe convective heating that can exceed material limits and directly influence structural integrity, TPS mass, and overall mission performance. The analyses in this report explain how heat flux and total heat load evolve over time and how the thermal environment differs across key regions of the vehicle. This forms a foundation for the subsequent TPS sizing efforts that complete the project.

Methodology

The trajectory data from the Phase 3 data sheet provides the time, altitude, and velocity throughout flight. These quantities are used to find freestream pressure, density, and temperature computed via an atmospheric model, `atmosphericCalculator`. The Mach number determines the modeling approach used for the shock layer. When $M_\infty \geq 5$, equilibrium air effects are included by applying an equilibrium chemistry solver, `equiflow`, to determine post-shock temperature, pressure, enthalpy, density, viscosity, and conductivity. When $M_\infty < 5$, a calorically perfect gas model is used instead, and the post-shock state is obtained from the standard normal shock relations. These relations supply the downstream conditions needed to compute the flow properties entering the stagnation region.

1.1 Stagnation Point Heating

Once the post-shock state is known, the stagnation point heating is evaluated using the appropriate high-speed heating model. For hypersonic conditions where $M_\infty \geq 5$, the analysis uses the Fay-Riddell stagnation point heat flux correlation,

$$\dot{q}_{FR} = 0.76 Pr^{-0.6} (\rho_e \mu_e)^{0.4} (\rho_w \mu_w)^{0.1} \sqrt{\frac{du}{dx}} (h_{0,e} - h_w), \quad (1.1)$$

where ρ_e , μ_e , and $h_{0,e}$ are post-shock properties, h_w is the wall enthalpy, and du/dx is the stagnation streamline velocity gradient. Wall viscosity μ_w is computed using Sutherland's law at the assumed cold-wall temperature. As the vehicle decelerates below Mach 5, the calculation transitions to the Sutton-Graves relation,

$$\dot{q}_{SG} = k \sqrt{\frac{\rho_\infty}{r_n}} U_\infty^3, \quad (1.2)$$

which provides a reliable estimate of convective heating in supersonic flow. When the Mach number drops below 1.2, heating is set to zero while time integration continues. The resulting heat flux history is integrated over time to produce the stagnation point heat load needed for later TPS evaluation.

1.2 Aerodynamic Heating

The second component concerns the aerodynamic heating at two midbody locations: one along the top surface and one along the side surface. These points experience finite flow turning angles, so the post-shock state is determined from oblique shock relations. For $M > 5$, the `equiflow` oblique shock function was used. For $2 < M < 5$, calorically perfect gas calculations were used, and for $M < 2$, aerodynamic heating was set to zero.

For the aerodynamic heating at the top and side midpoints, the nodal geometry is imported directly from the data sheet. The coordinates are converted into a surface distance s from the nose to each midpoint by summing the segment lengths between successive nodes. This distance provides the characteristic length scale required for computing the length-based Reynolds number.

The post-shock state at each location is then computed using either the equilibrium oblique shock solver for $M_\infty \geq 5$ or the calorically perfect gas oblique-shock model for $M_\infty < 5$. The equilibrium solver provides all transport properties. In the calorically perfect gas region, viscosity is again assigned using Sutherland's law and conductivity by assuming a constant Prandtl number.

Once the post-shock state is known, the aerodynamic heating is computed using the reference temperature method. The film properties ρ^* , μ^* , k^* , and C_p^* are evaluated at T^* using the EquiFlow property solver. The reference Reynolds number and critical Reynolds number are then calculated. If $Re_s < Re_{cr}$, laminar heating correlations are used. If $Re_s > Re_{cr}$, turbulent correlations are used. The convective heat flux and recovery enthalpy are then found. The resulting heat flux at each point is integrated with respect to time using the trapezoidal rule to produce the total heat load at the top-midpoint and side-midpoint locations.

1.3 TPS Sizing

All calculations for the TPS sizing were performed using the approximate ablative TPS finite difference code executable provided by Dr. Dec. The margin policy was applied to two separate materials to find the required thickness: Carbon-Carbon and a fake PICA. First, the `bipara.txt` file was modified to include time, density, pressure, velocity, heat flux, and heat load. These values were all calculated in previous parts of Phase 3. Next, in the command prompt window, `PESST_FD.exe` was used to find the required TPS thicknesses.

Per the margin policy, three cases needed to be examined for each material. In Case 1, heat flux is multiplied by the trajectory dispersion factor of 1.2. In Case 2, heat flux is multiplied by the same trajectory dispersion factor as well as an aerodynamic heating uncertainty of 1.25, totaling an overall factor of 1.5. In Case 3, the bond line temperature is

reduced by 55.6 degrees Celsius and then modified by only the trajectory dispersion factor of 1.2.

Results and Discussion

2.1 Stagnation Point Heating

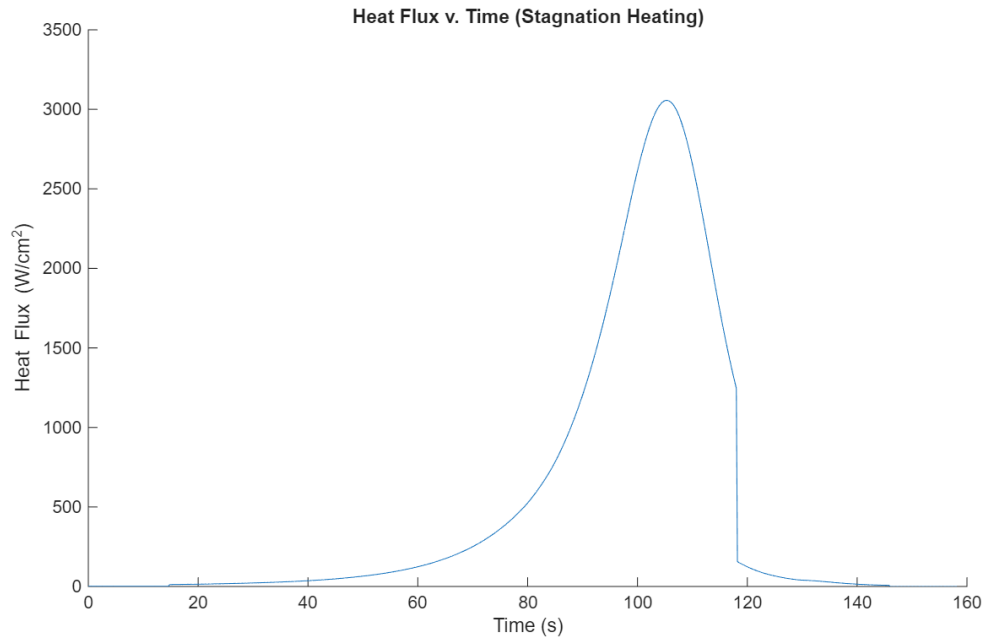


Figure 2.1: Stagnation point heat flux as a function of time.

The stagnation point heat flux history shows how the vehicle's thermal environment evolves as it accelerates through hypersonic flow and later descends into denser atmosphere. At early times, the heating rate remains very low because the vehicle is still at high altitude with relatively low dynamic pressure. As the velocity increases and the trajectory brings the vehicle into denser air, the stagnation point heating rises sharply. The peak occurs shortly after 100 seconds, where the combination of high velocity and significantly increased freestream density produces the maximum convective load. This behavior is consistent with both the Fay-Riddell and Sutton-Graves correlations, which scale strongly with velocity and density. Following the peak, the heat flux rapidly decreases as the vehicle decelerates and passes below Mach 5, after which the stagnation heating drops to zero once the Mach number falls below 1.2.

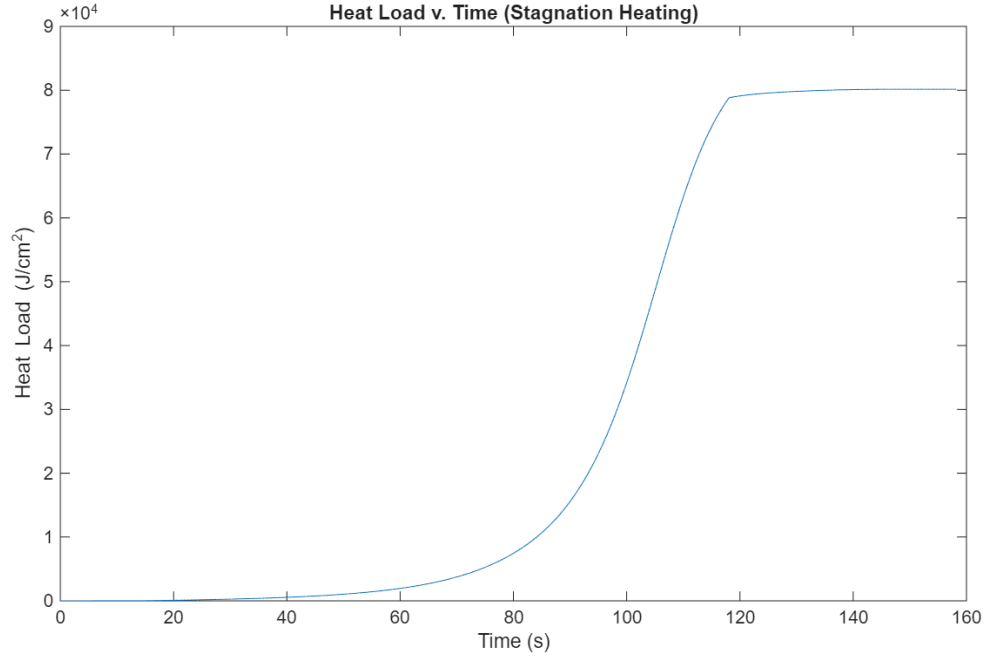


Figure 2.2: Stagnation point heat load as a function of time.

The cumulative stagnation point heat load increases slowly in the early part of the trajectory and then grows rapidly during the high-heating portion, reflecting the same trend seen in the heat flux curve. After the peak heating period ends, the heat load curve begins to level off because additional heating becomes negligible. The final heat load value represents the total energy absorbed at the stagnation point over the entire flight segment and is a key input for the thermal protection system sizing software.

2.2 Aerodynamic Heating

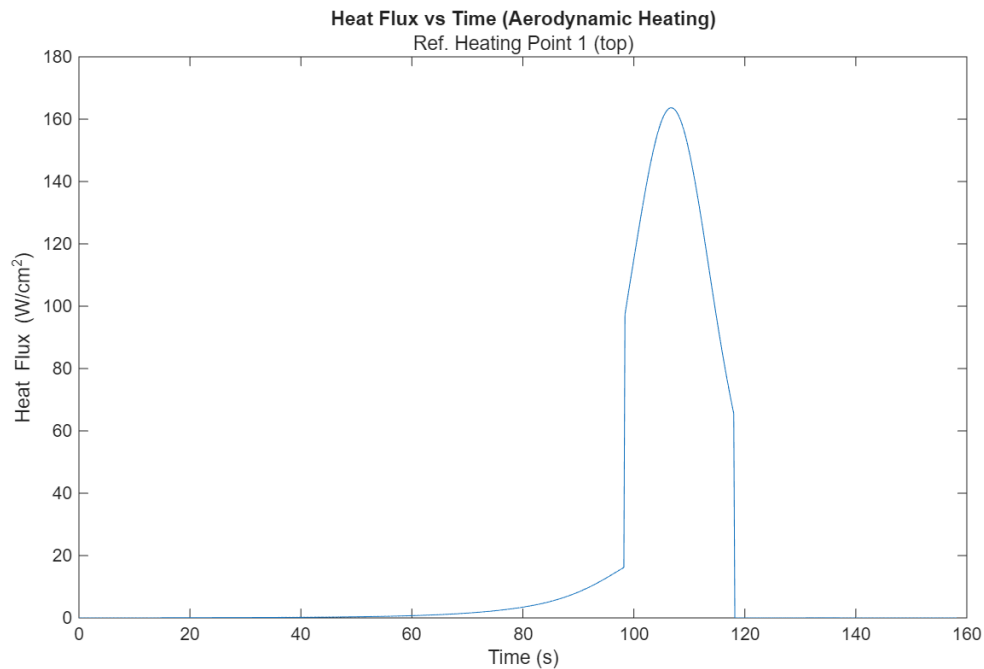


Figure 2.3: Aerodynamic heat flux at reference heating Point 1 as a function of time.

The aerodynamic heating histories at the two varied locations show the same overall structure as the stagnation point heating but with significantly reduced magnitudes due to the presence of oblique rather than normal shocks. At the top midpoint, Point 1, the heat flux remains very small through the early trajectory until the vehicle begins to descend and the post-shock temperature and velocity increase. The heating rate then rises rapidly, reaching its peak shortly after 100 seconds. This behavior reflects the strong dependence of the reference temperature Stanton number on both velocity and the length-based Reynolds number, which increases sharply as the vehicle enters denser atmosphere. Once the Mach number drops and the oblique shock strength weakens, the heat flux decreases quickly and becomes negligible, consistent with expected aerodynamic heating trends for surface locations away from the stagnation region.

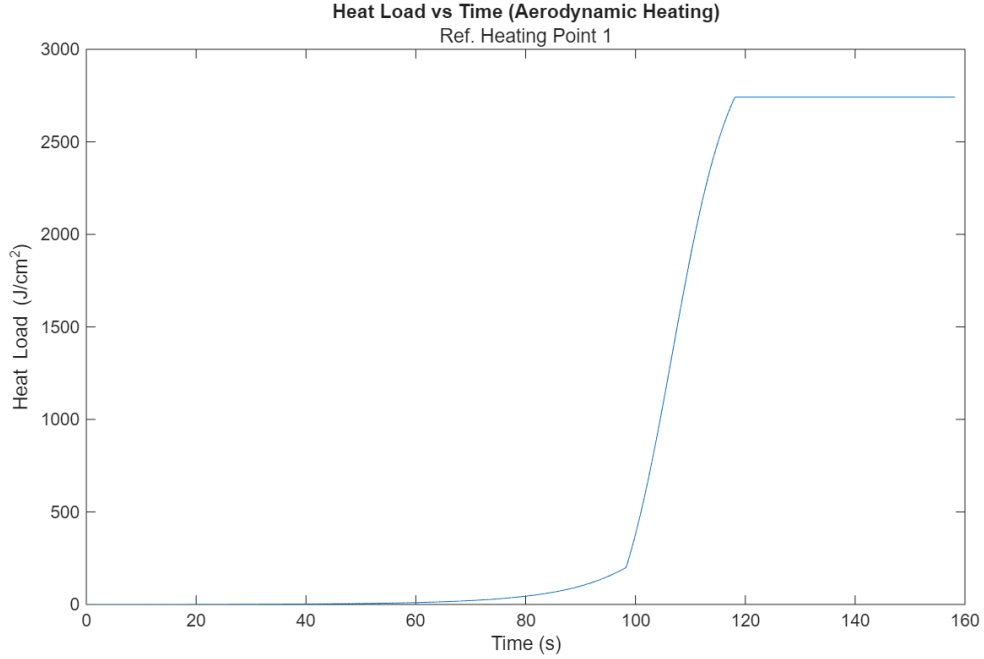


Figure 2.4: Aerodynamic heat load at reference heating Point 1 as a function of time.

The top midpoint heat load graph follows the same pattern. There is a slow accumulation in the early portion of the trajectory followed by a steep rise during the high-heating window and a plateau once the heating rate approaches zero. The final heat load is significantly lower than the stagnation point value, demonstrating the reduced thermal severity on the upper surface. This is consistent with the flow turning angle on the top surface, which produces a weaker post-shock state than at the stagnation line.

At the side midpoint, Point 2, the heating behavior is similar in timing but slightly higher in magnitude. The heat flux peak is larger than at the top-mounted point, which indicates stronger compression and higher edge enthalpy on the side surface. This difference is expected because the side surface flow turning angle is larger, producing a stronger oblique shock and a higher post-shock temperature. The heat load curve for this point shows the same rapid rise and eventual flattening but reaches a higher total heat load than the top midpoint. This shows that an increase of the cone angle would directly increase the heat flux on the cone section.

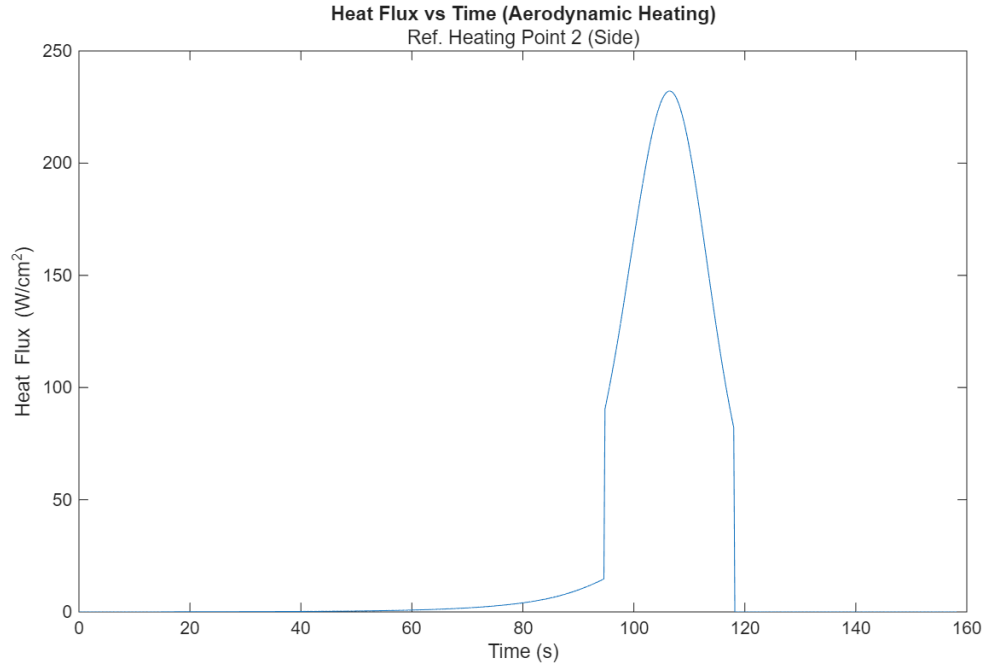


Figure 2.5: Aerodynamic heat flux at reference heating Point 2 as a function of time.

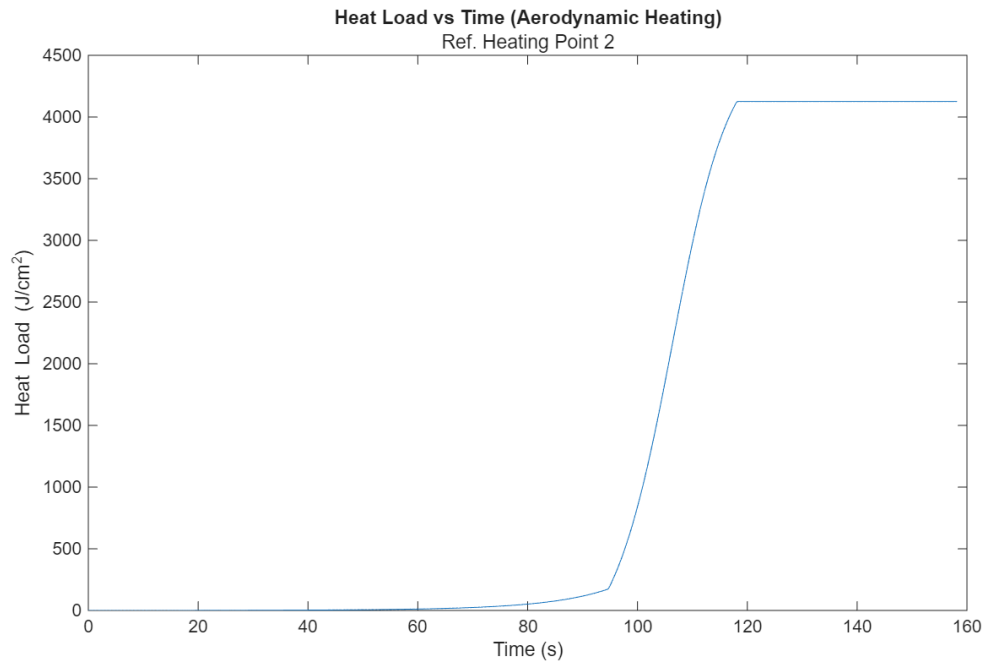


Figure 2.6: Aerodynamic heat load at reference heating Point 2 as a function of time.

2.3 TPS Sizing

Table 2.1: Carbon-Carbon TPS Sizing Results

Parameter	Case 1	Case 2	Case 3
CC thickness (cm)	0.3175	0.3175	0.3175
PolyFoam (cm)	7.3803	7.5321	7.8218
Total recession (cm)	3.0534	3.8441	3.0534
Total thickness (cm)	10.7512	11.6937	11.1927
Areal weight (kg/m ²)	170.712	187.858	177.113

Table 2.2: Fake PICA TPS Sizing Results

Parameter	Case 1	Case 2	Case 3
Calculated TPS thickness (cm)	34.1324	42.1053	34.2917
Thickness required for insulation (cm)	3.5274	3.5871	3.6867
Total recession (cm)	30.6050	38.5182	30.6050
Areal weight (kg/m ²)	77.635	95.770	77.998

The fake PICA thermal protection system behaves very differently from the Carbon-Carbon configuration. Although PICA materials experience substantially larger surface recession, they are far less dense than a Carbon-Carbon TPS. This tradeoff results in the PICA material weighing less than half the CC material, despite being three to four times thicker.

Among the applied margin factors, the aerodynamic heating uncertainty has the greatest influence on TPS sizing. Since convective heat flux appears directly in both the ablation and conduction formulations, a 25 percent increase in heating substantially increases recession depth and steepens the temperature gradient. The bondline temperature margin contributes additional insulation thickness but does not affect surface recession, making its impact less significant for ablation-dominated materials such as the fake PICA.

The initial TPS temperature also affects the required thickness. A higher starting temperature reduces the allowable thermal margin before reaching the 250°C bondline limit, leading to greater insulation requirements. This effect is more pronounced in the Carbon-Carbon system, which relies heavily on internal insulation layers. Increasing the initial temperature directly increases the necessary foam thickness. In contrast, the fake PICA system is influenced less by initial temperature because its total thickness is governed primarily by ablative recession, which depends more strongly on convective heating and surface energy balance than on internal conduction.

Appendix A: Instructions on Running Code

The overhead function is `aeroThermoPPthree`. However, to make this code more efficient, two functions were created in their own scripts whose outputs are workspaces that become the inputs to the overhead function.

The script `preStagHeating` has a `parfor` loop in it that takes approximately five minutes to run and outputs the workspace `StagHeatingWS`. Next, `preAeroHeating` also has a `parfor` loop that takes approximately five to eight minutes to run and eventually outputs the workspace `aeroHeatingWS`. Running either of those scripts requires the Parallel Computing Toolbox. Both workspaces were included in the submitted zip file for convenience.

Once the two workspaces are available, call the overhead function as follows:

```
aeroThermoPPthree('StagHeatingWS', 'aeroHeatingWS')
```

This automatically runs the other two functions included in the zip file, `stagHeating` and `aeroHeating`, and then outputs the plots. The version of `atmosphericCalculator` included in the zip file is slightly modified from the one provided by Dr. Dec. The only input is altitude, and velocity is no longer an output.